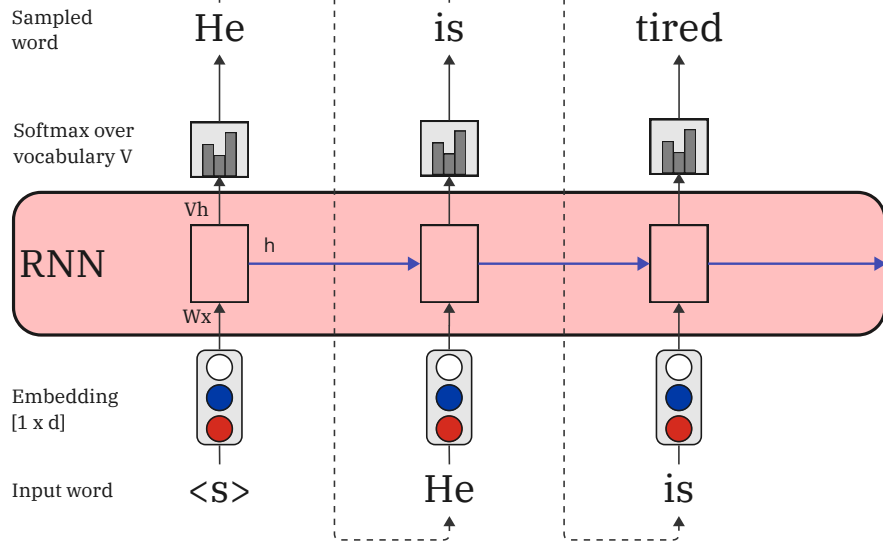




$$P(\text{tired} \mid \text{is}) = \arg \max_i y_{t,i}$$

$$y_t = \text{softmax}(Vh_t)$$

$$h_t = g(Uh_{t-1} + Wx_t)$$

RNN-based LMs by definition take into account preceding tokens in a sequence via recurrent connections.

Examples (generation): Seq2seq: Sutskever et al. (2014), Attention: Bahdanau et al. (2015).

Examples (embeddings): ELMo (2018), ULMFiT (2018)